

Computer Science & Information Technology

Computer Network 2

Network Layer

DPP: 1

- Q1** In IPv4 packet header, identification number field is used _____.
 (A) To prevent indefinite traversing of IP packets.
 (B) To identify IP fragments from same original segment.
 (C) To identify sequence of IP fragments of same original segment.
 (D) None of the above
- Q2** Consider Host A sends a UDP packet to Host B, but Host A receives ICMP "Destination Unreachable" message in response to the packet. Which of the following scenarios could lead to this ICMP message being generated?
 (A) The TTL (Time-to-Live) value reaches zero before reaching Host B.
 (B) The port number used is currently closed on Host B.
 (C) Host B IP address changed dynamically.
 (D) Host B is disconnected from the network.
- Q3** Supernetting is a process of _____.
 (A) Dividing a network into smaller sub-networks
 (B) Combine multiple networks into a larger one
 (C) Increasing the number of hosts per subnet
 (D) Reducing the number of hosts per subnet
- Q4** Consider following network addresses present in the router's routing table.
 170.180.190.64 / 27
 170.180.190.96 / 28
 170.180.190.112 / 28
 What can be the supenet address?
 (A) 170.180.190.64 / 25
 (B) 170.180.190.64 / 26
 (C) 170.180.190.64 / 27
 (D) Supernetting is not possible in given network addresses
- Q5** NAT (Network Address Translation) is used for _____.
 (A) Dynamic IP address allocation
 (B) Creation of subnets within a network
 (C) Multiple devices to share a single public IP address
 (D) Find MAC address corresponds to given IP address
- Q6** Consider a host uses 'token bucket' algorithm for traffic_shaping. The capacity of the token bucket is 100 Megabytes and initially filled to its capacity. The token generation (arrival) rate is 20 Megabytes per seconds and the bottleneck bandwidth of the transmission link is 30 Megabytes per seconds. What is the maximum duration (in seconds, closest to the nearest integer) for which the computer can transmit at maximum bandwidth to handle burst traffic?
- Q7** The DHCP client first sent _____ DHCP message to locate available DHCP servers ?
 (A) DHCPDISCOVER (B) DHCPOFFER
 (C) DHCPREQUEST (D) DHCPACK
- Q8** Consider a router has three full-duplex 100 Mbps interface and minimum frame size is 3000 bits (including all overheads). What is the maximum packet processing time (in microseconds) at router for wirespeed forwarding?



Answer Key

Q1 B
Q2 B, C, D
Q3 B
Q4 B

Q5 C
Q6 10~10
Q7 A
Q8 10~10



[Android App](#)



[iOS App](#)



[PW Website](#)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

- (a) TTL
- (b) True
- (c) Offset

Q2 Text Solution:

- (A) Time Exceeded
- (B) Port Unreachable
- (C) Host Unreachable
- (D) Host Unreachable

Q3 Text Solution:

- (A) Subnetting
- (B) Supernetting

Q4 Text Solution:

170.180.190.96 / 28 170.180.190.01100000 / 28

170.180.190.112 / 28 170.180.190.01110000 / 28

After one-bit supernetting

Supernet Add1 170.180.190.01100000 / 27

170.180.190.64 / 27 170.180.190.01000000 / 27

After one-bit upnetting

Supernet Address 170.180.190.01000000 / 26

170.180.190.64 / 26

Q5 Text Solution:

- (A) DHCP
- (B) Subnetting
- (C) NAT
- (D) ARP

Q6 Text Solution:

C = 100 MB (Full)

R = 20 MB/sec

M = 30 MB/sec

S = ?

C + RS = MS

$$S = \frac{C}{(M-R)}$$

$$S = \frac{100 \text{ MB}}{(30 \text{ MB/sec} - 20 \text{ MB/sec})}$$

$$S = \frac{100 \text{ MB}}{10 \text{ MB/sec}} = 10 \text{ sec}$$

Q7 Text Solution:

Time Required to receive one minimum size frame

$$= \frac{\text{minimum frame size}}{\text{bandwidth}} = \frac{300 \text{ bits}}{100 \text{ Mbps}} = 30 \mu\text{s}$$

Maximum packet processing time

$$= \frac{30 \mu\text{s}}{3} = 10 \mu\text{s}$$



[Android App](#)

| [iOS App](#)

| [PW Website](#)